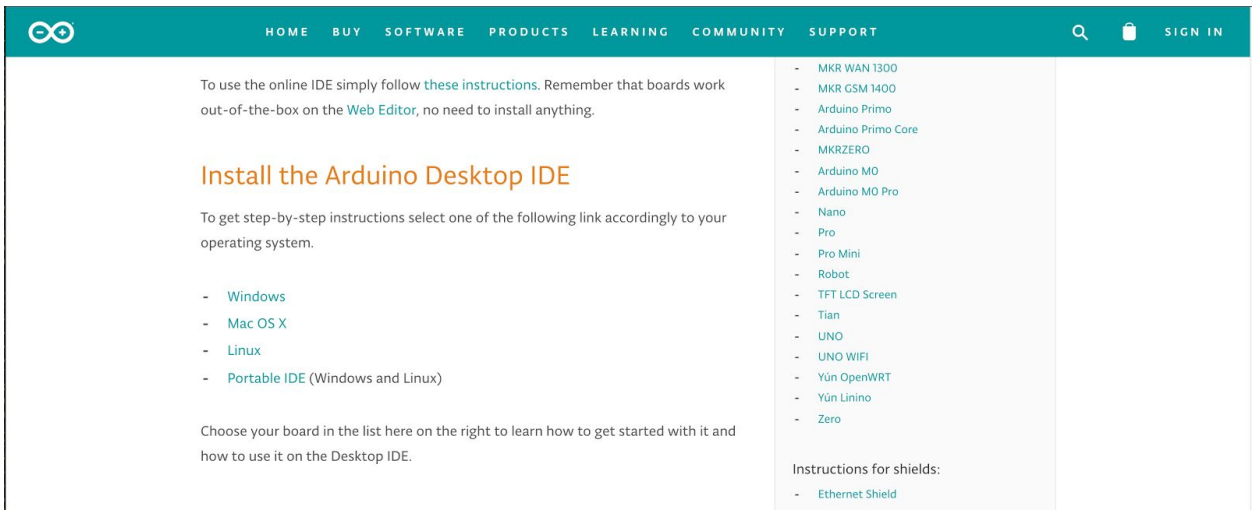
Loading the Infrared Proximity Sensor Firmware to an Arduino Uno Board

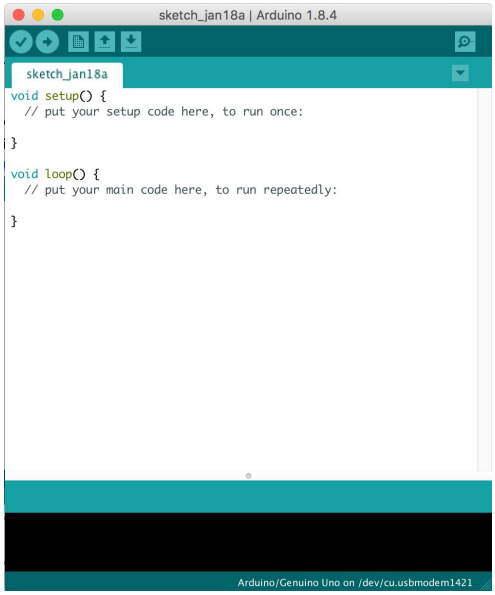
Please see [Setting Up the Infrared Proximity Sensor with an Arduino Uno Board] *before proceeding with this tutorial.*

1. Installing the Arduino Desktop IDE

You will need the **Arduino Desktop IDE** to run the proximity sensor source code. [Follow this guide] and choose your operating system to install the **Arduino Desktop IDE**.



Once the **Arduino IDE** has finished installing, open it. It will look like this:

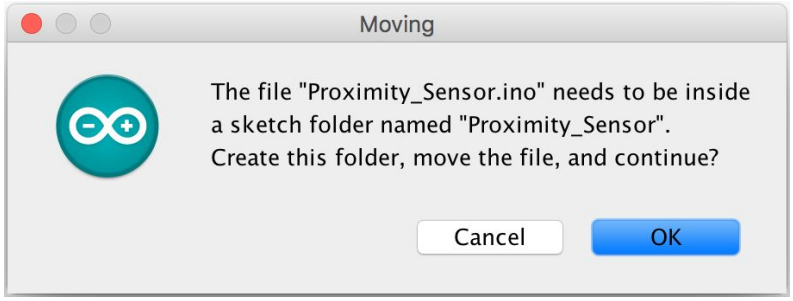


2. Downloading and opening the source code

To download the source code, right click this link and click on **'Save Link As...'** to save the ***Proximity_Sensor.ino*** file to the folder of your choice.

Open ***Proximity_Sensor.ino*** in the **Arduino IDE**. You can do this by going to the top navigation bar of the IDE. Click on **File > Open**. Navigate to the folder where you stored ***Proximity_Sensor.ino*** and click on the file to open it.

A pop-up window with a message about creating a sketch folder (shown below) will appear. Click 'OK'. This will create the needed sketch folder in which the ***Proximity_Sensor.ino*** file will be located.



3. Connecting the Arduino Uno Board to the Arduino IDE

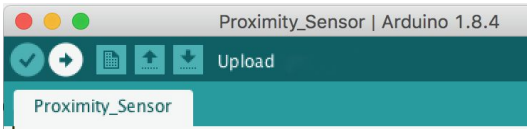
Follow the steps in this tutorial to connect the Arduino Uno board to the **Arduino IDE**. This guide will assist you with selecting the Arduino board type and port within the **Arduino IDE**.

If you are using a Macintosh or Linux operating system, see this tutorial to find the Arduino port.

4. Uploading the Sensor Source Code to the Arduino Uno Board

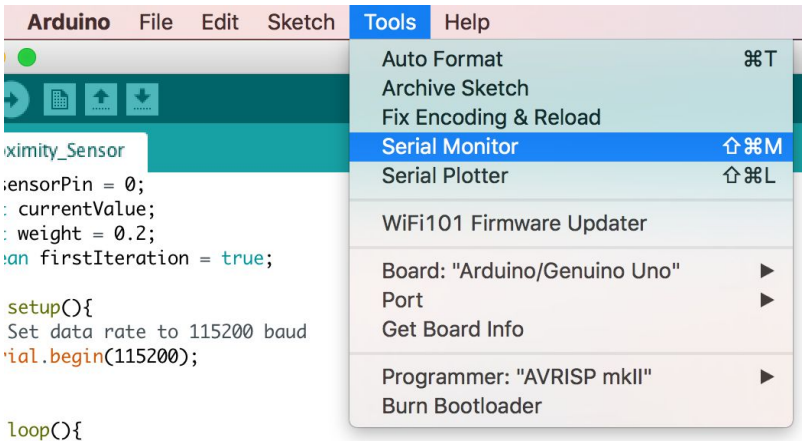
Once the board type and port are selected, you are ready to begin uploading the source code to the **Arduino Uno board**.

In the top left corner of the **Arduino IDE** window, click the **Upload** button (the button that looks like an arrow).

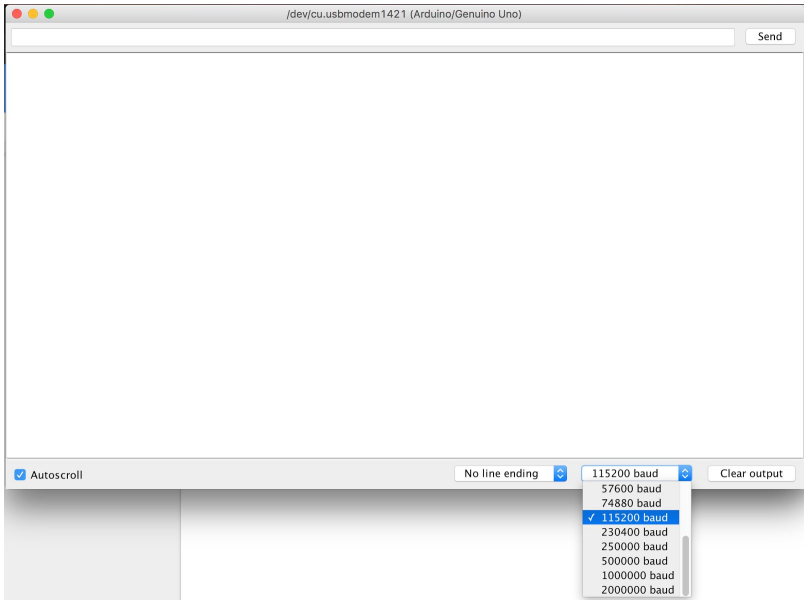


The status bar near the bottom of the window will say **'Done uploading'** when the code has finished uploading to the board.

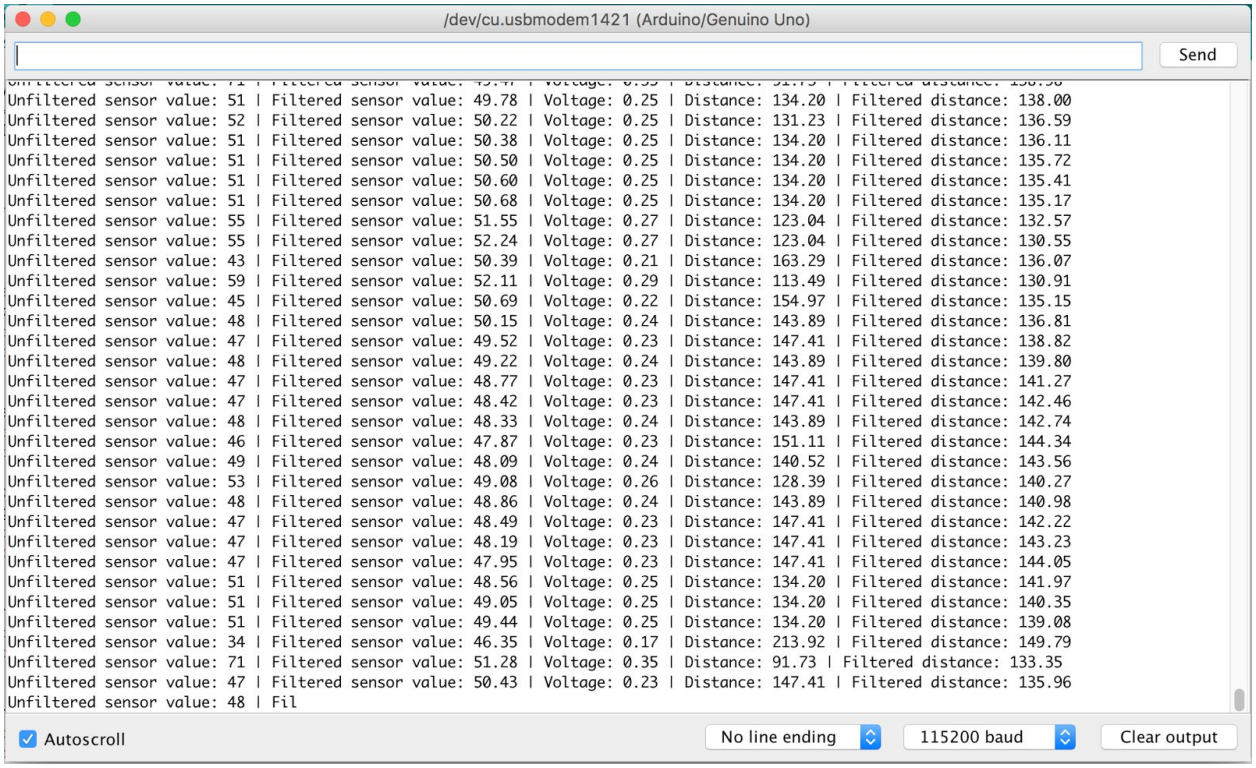
To view the data from the sensor, you will need to open the Serial Monitor. At the very top navigation bar, click **Tools** > **Serial Monitor.**



Once the Serial Monitor window is open, make sure that **115200 baud** is selected from the drop-down box at the bottom of the window.



You will then be able to see the data from the sensor appear on the Serial Monitor.



Because the analog data values coming from the sensor have some noise, a low-pass filter has been implemented in order to smooth out the raw sensor values.

The **unfiltered sensor value** is the raw sensor value (without the smoothing filter applied to it).
The **filtered sensor value** is the sensor value in which the smoothing filter has been applied.
The **voltage** is the analog voltage based on the unfiltered sensor value.
The **distance** is the distance of an object from the sensor, based on the unfiltered sensor value (measured in centimeters).
The **filtered distance** is the distance of an object from the sensor, based on the filtered sensor value (measured in centimeters).